FirstLight

PITCH DECK

“First light in the darkest hour.”

Our Best Team

We are a team passionate about building AI-powered drone systems to transform disaster response. By integrating advanced AI orchestration with autonomous swarms, we aim to provide first responders with **“FirstLight”** — a reliable eye in the sky that operates tirelessly, autonomously, and safely in environments where human access is limited.

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Case Study 3

Decentralized Swarm Intelligence

Problem Statement

The Challenge: The 72-Hour Blackout

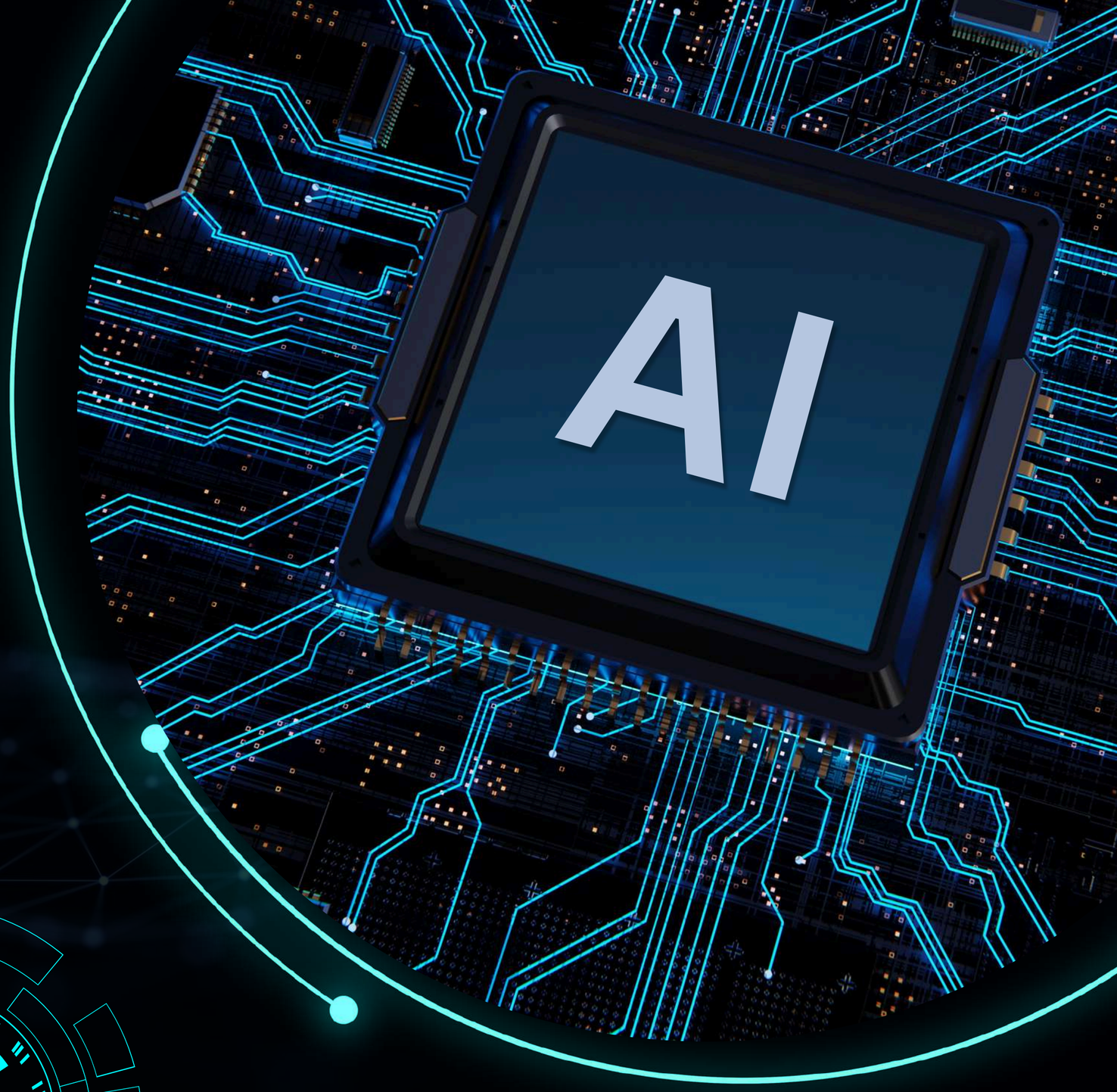
- **ASEAN Vulnerability:** High-frequency typhoons and earthquakes.
- **Critical Failure:** Standard cloud-AI collapses during communication blackouts.
- **The "Golden Window":** Life-saving response is stalled by zero connectivity.

SDG Alignment

- **SDG 3: Good Health & Well-being**
  - Target 3.d: Ensuring timely disaster response and medical aid to survivors.
- **SDG 9: Industry, Innovation, & Infrastructure**
  - Target 9.1 & 9.5: Improving resilient emergency infrastructure and applying advanced innovation in disaster response.

Proposed Solution

- Autonomous, Decentralized Command Agent
  - Converts mission goals into reasoned, sequenced actions (movement, relaying, and recall) without human intervention.
- MCP Tool Layer for Standardized Control
  - Uses structured schemas to transform drone capabilities into "discoverable tools" for dynamic, plug-and-play mission planning.
- Self-Healing Swarm Execution
  - Maintains operations during blackouts via local autonomy—managing battery, handoffs, and reconnection maneuvers.



Demo

Technologies Used

AI & Orchestration Engine:

Google Vertex AI (vertexai) running state-of-the-art LLMs (Gemini 2.5) to act as the cognitive engine for autonomous mission planning and multi-agent coordination.

Protocol & Backend Services:

Node.js + Express + TypeScript functioning as the FirstLight MCP Server. This acts as the bridge that standardizes communication between the AI orchestrator and the simulated drone components.

Mission MCP Tools Modules:

Specialized MCP tools modules for drone control, scan intelligence, communication mesh status, relay operations, and orchestration policies.

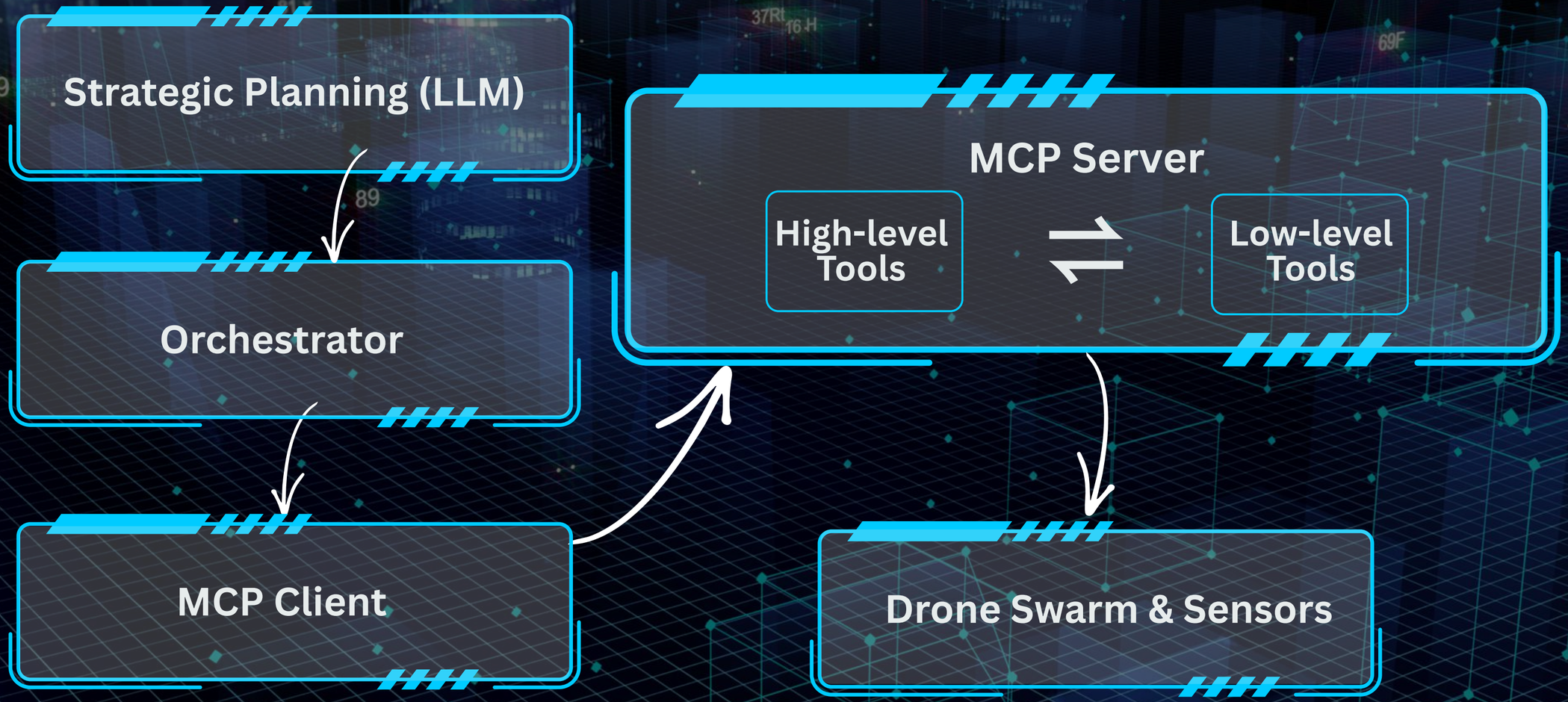
Frontend simulation and dashboard:

React + TypeScript + Vite, with interactive mission UI and operator chat/control panels.

Geospatial & 3D Visualization:

CesiumJS (3D globe simulation), Deck.gl (Data overlays), MapLibre-gl (Mapping engine) and Three.js (Drone & FPV rendering) are deeply integrated for rendering a realistic 3D simulation map and providing dynamic FPV (First-Person View) Drone Cams.

MCP Architecture



Innovation

Geospatial Predictive Nature

- Initial survivor probability is calculated by fusing **Building Density**, **Residential Factors**, and **Road Access**
- Instead of starting with a blank map, our model focuses early search efforts on **high-density residential areas** to speed up the searching process.

Blanket Search

- **Wide Scan** - The default mode, which detects initial signs of survivors for **rapid area coverage**.
- **Micro Scan** - Automatically activated when a drone detects a high signal probability, then performs repeated, high-precision scans to **investigate the signa**l.

Innovation

Collaborative & Adaptive Sensor

- **Various sensors** are taken into consideration to compute the probability accurately. (Mobile, Thermal, Wi-Fi, Sound)
- In Micro Scan mode, drone sensors' confidence **builds up over time** to prevent false positives due to noise.
- As survivors are found, the AI **"learns"** which sensors are providing the most reliable data and **adjusts their weights**.

Global Perspective of Orchestrator

- Raw telemetry is compressed into a semantic **Natural Language Summary**, allowing the AI to read the battlefield and make strategic decisions like a human commander.
- Every 10 ticks, the entire grid and drone status are passed to AI, ensuring the AI **understands the exact state** of every drone, battery level, and scanned cell simultaneously.

Innovation

Resource Management

- Before a drone accepts an assignment, the system runs a **Battery Forecast** to ensure it has enough energy to reach the target and **return to base to charge** when it reaches the safety threshold.
- When a drone's battery is low, it **negotiates task handoff** with the nearest available drone, passing its high-probability target coordinates so the investigation continues without interruption.

Information Time-To-Live

- As time passes, the **confidence level** in a previous negative result for the area will **decay**, triggering **re-verification** of the area.
- Hence, the swarm now understands that a disaster zone is a **changing environment**, preventing missing out on any survivors.

Innovation

Self Healing Communication Bridge

- The system implements an **Autonomous Relay Rotation** where a backup relay at base replaces a low-battery relay location in the field without dropping connectivity for a single tick.
- If any drones are **almost losing connection**, the relay will **move toward the centroid** of all drones to maximize drone connectivity.

Network Connectivity Resilience

- If a drone loses signal for a certain period, it will **fall back** towards the region within the base station connection.
- Offline drones continue scanning and store data in a local **Offline Buffer**, which is "flushed" to the base the moment a connection is restored, ensuring **zero data loss**.

Business Model

1. GOVERNMENT & PUBLIC SAFETY (B2G LICENSING)

- **Annual Platform Licenses:** Tiered access for National Disaster Agencies (e.g., FEMA, ASEAN AHA Centre) and Civil Defense units.
- **SOP Integration & Onboarding:** One-time setup fees for country-specific hazard mapping, language localization and local agency protocol integration.
- **Simulation as a Service:** Recurring subscriptions for AI-driven preparedness drills and digital-twin disaster scenarios.

2. ENTERPRISE & INDUSTRIAL RESILIENCE (B2B)

- **Infrastructure Resilience Modules:** Emergency response AI designed to handle power grid failures and industrial accidents, ensuring continuous operations and business continuity.
- **PaaS (Platform as a Service):** A subscription tier for private responders and NGOs providing real-time probability heatmap APIs and updated MCP tool endpoints.
- **Systems Integration:** Customizing the FirstLight MCP server to interface with bespoke legacy drone fleets or existing monitoring system

Market Segment

Public Sector: Government, NGOs

- **Use Case:** Rapid area scanning and survivor localization in the "Golden 72 Hours" post-typhoon or earthquake.
- **Standout Point:** Swarm-based coverage is 10x faster than manual teams, generating real-time probability heatmaps for high-speed decision-making.

Private Sector: Search & Rescue (S&R) Service Providers

- **Use Case:** Missing persons in remote terrain (forests/mountains) and reconnaissance for industrial collapses or hazardous sites.
- **Standout Point:** Replaces slow manual grid searches with AI-driven prioritization, significantly reducing operational risks for human rescuers.

Humanitarian: Humanitarian Aid Groups

- **Use Case:** Urban Search & Rescue (USAR) following airstrikes or infrastructure failure in GPS-denied/hostile environments.
- **Standout Point:** Decentralized autonomy makes the system robust against jamming and signal loss — prioritizing life-saving search where infrastructure is non-existent.

Competitor Analysis

Traditional Drone Systems

Strengths

- Strong hardware & reliable flight systems
- Mature ecosystem

Gaps

- Human-dependent operation
- Requires continuous communication
- Limited autonomy in blackout scenarios

Autonomous Drone Software

Strengths

- Advanced navigation & perception
- High level autonomy

Gaps

- Limited multi-agent coordination
- Weak decentralized orchestration
- Limited scalability

Defense-Grade Platforms

Strengths

- Highly advanced and robust
- High-end capabilities

Gaps

- Extremely high cost
- Not scalable for developing regions

FirstLight

Strengths

- Dynamic drone discovery
- AI decision-making and reasoning
- Battery & relay aware
- Self-healing under failure
- Cost-effective deployment

Future Improvement

Sim-to-Real Edge Deployment (Local LLMs)

Transitioning from cloud-based models (Vertex AI) to hyper-efficient, open-weights models (Ollama) running completely locally on edge compute nodes inside the disaster zone. This completely severs the reliance on the cloud.

Multi-region scaling

Support simultaneous disaster zones with federated swarm coordination across provinces or countries.

Higher-fidelity simulation

Integrate weather, terrain obstruction, aftershock dynamics, and probabilistic sensor failure to improve realism.

Thank You